The 2D Fourier Transform

- Represent function on a new basis
  - Think of functions as vectors, with many components
  - We now apply a linear transformation to transform the basis
    - dot product with each basis element
- In the expression, u and v select the basis element, so a function of x and y becomes a function of u and v
- basis elements have the form

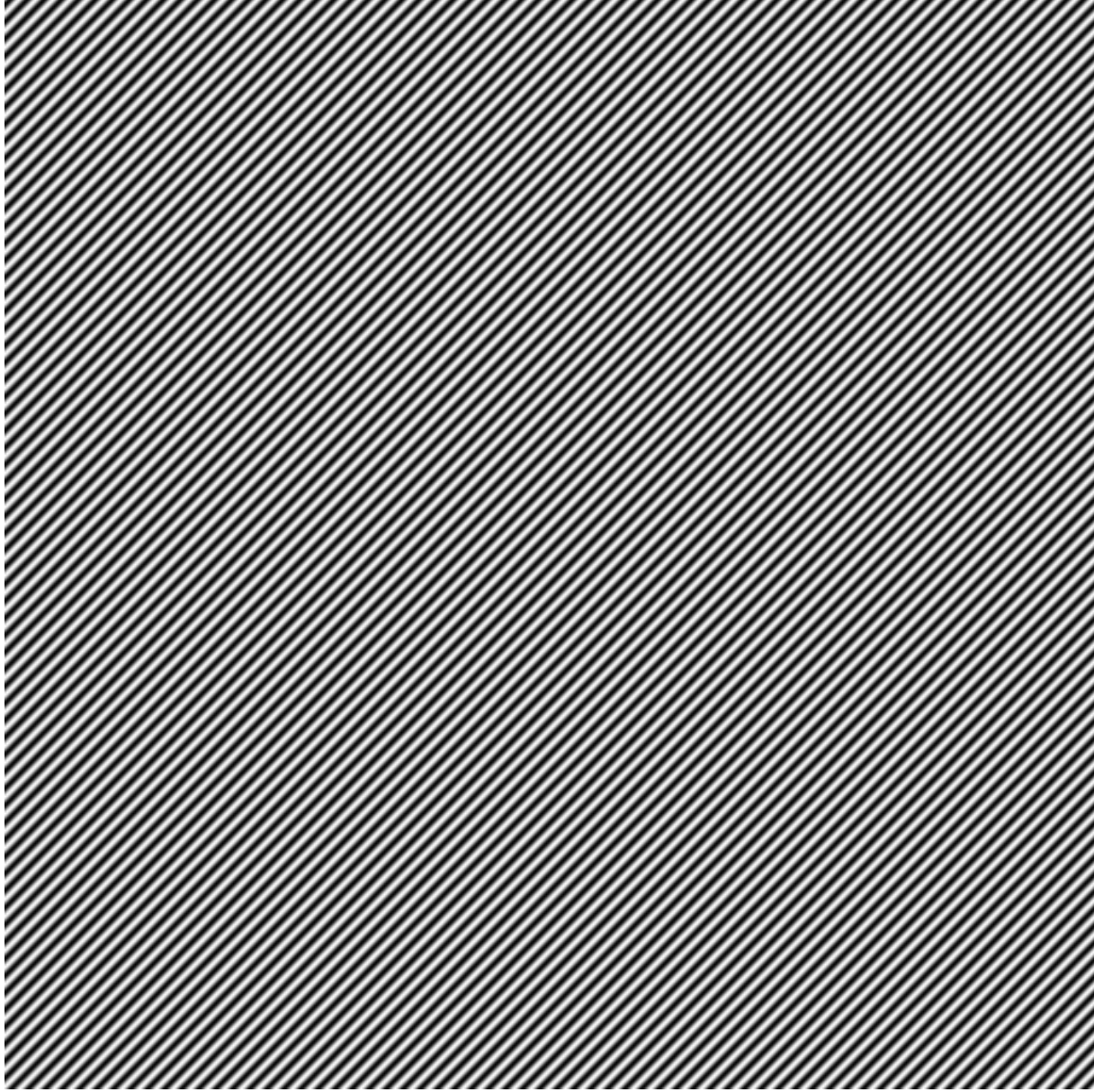
$$e^{-i2\pi(ux+vy)}$$

$$F(g(x,y))(u,v) = \iint_{\mathrm{R}^2} g(x,y) e^{-i2\pi(ux+vy)} dxdy$$

To get some sense of what basis elements look like, we plot a basis element --- or rather, its real part --- as a function of x,y for some fixed u, v. We get a function that is constant when (ux+vy) is constant. The magnitude of the vector (u, v) gives a frequency, and its direction gives an orientation. The function is a sinusoid with this frequency along the direction, and constant perpendicular to the direction.

Here u and v are larger than in the previous slide.

And larger still...



Phase and Magnitude

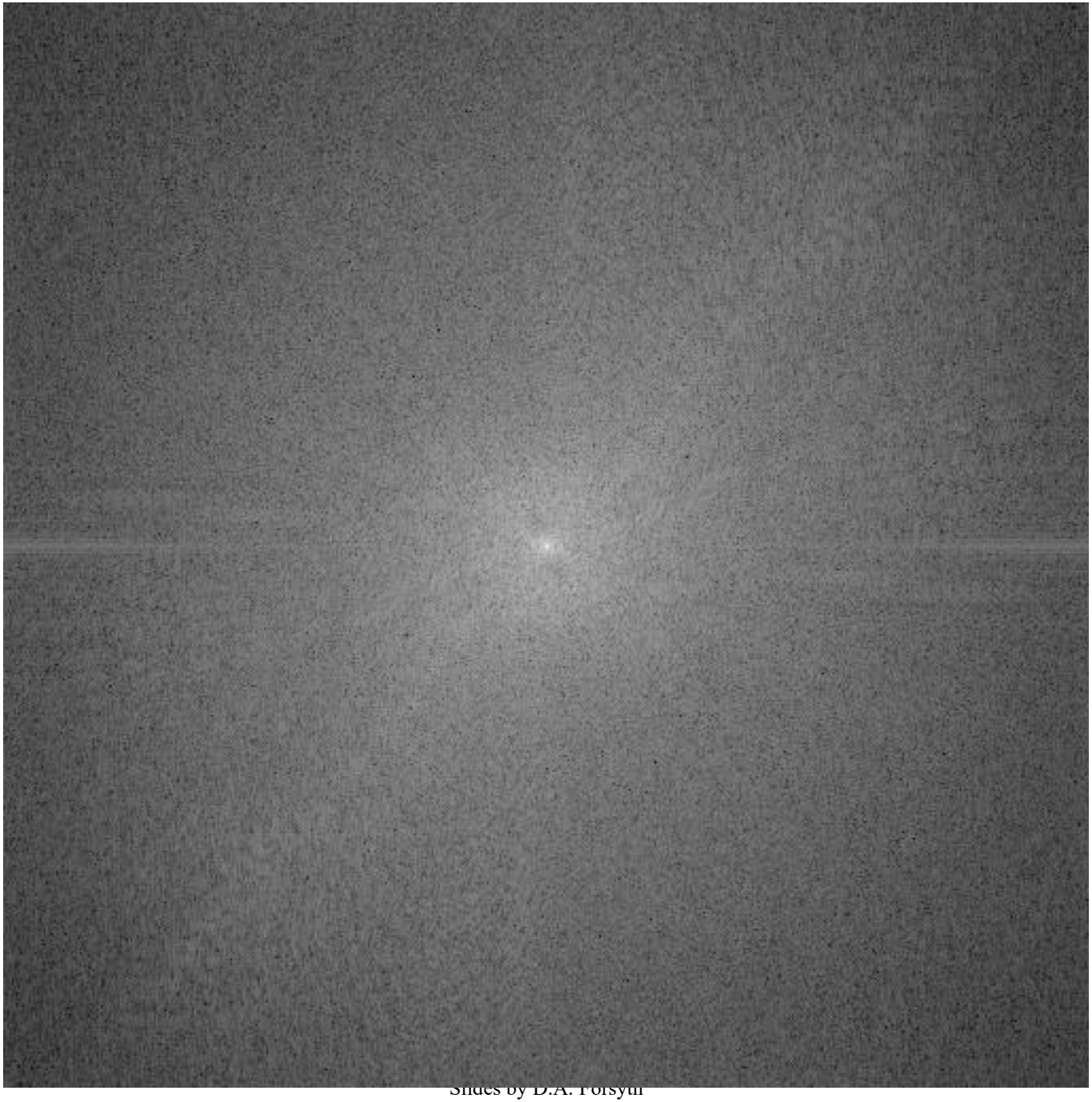
- Fourier transform of a real function is complex
  - difficult to plot, visualize
  - instead, we can think of the phase and magnitude of the transform
- Phase is the phase of the complex transform
- Magnitude is the magnitude of the complex transform

- Curious fact
  - all natural images have about the same magnitude transform
  - hence, phase seems to matter, but magnitude largely doesn't
- Demonstration
  - Take two pictures, swap the phase transforms, compute the inverse - what does the result look like?

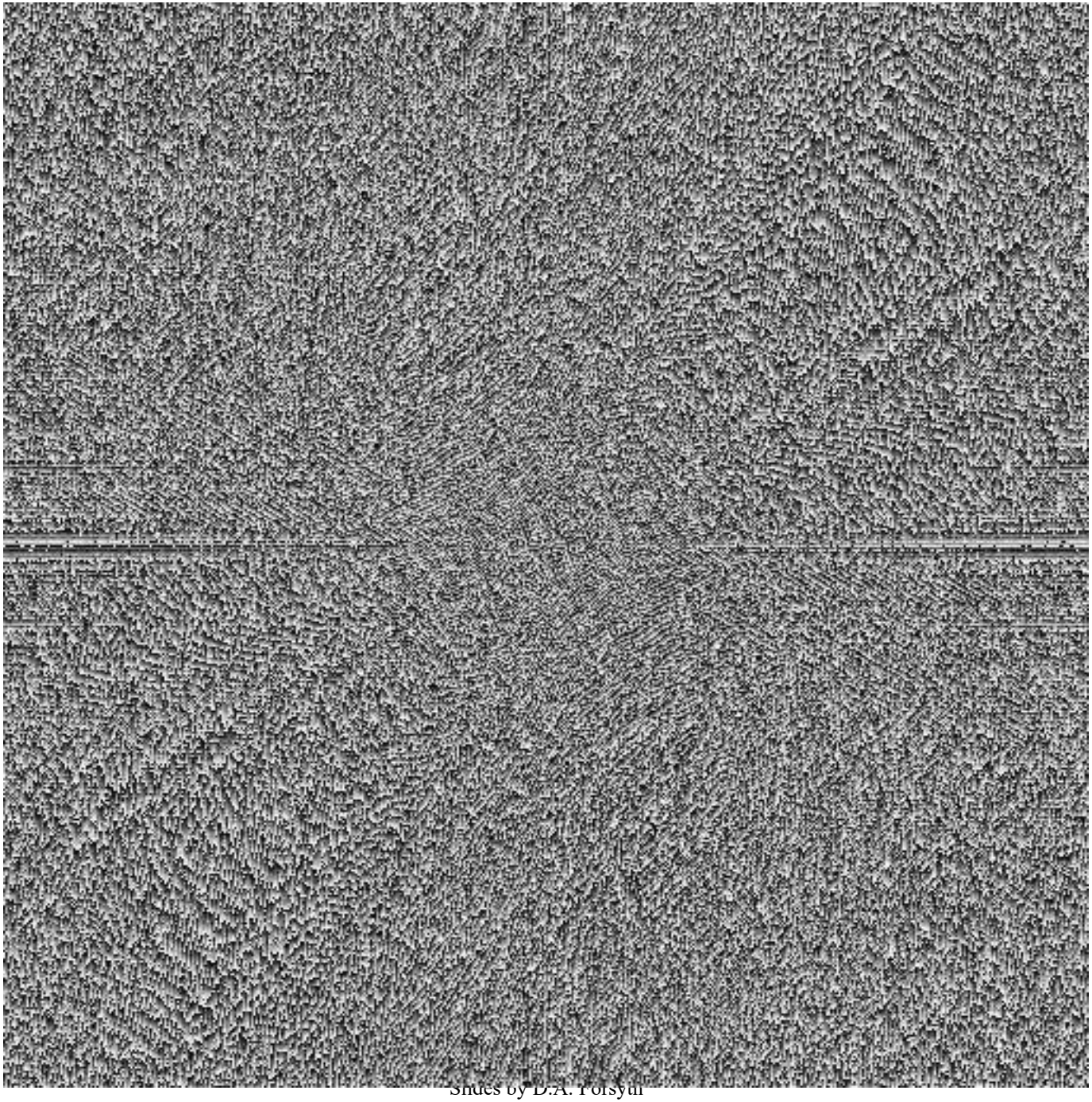
Read:
https://openaccess.thecvf.com/content_CVPR_2020/papers/Yang_FDA_Fourier_Domain_Adaptation_for_Semantic_Segmentation_CVPR_2020_paper.pdf

Slides by D.A. Forsyth

This is the magnitude transform of the cheetah pic



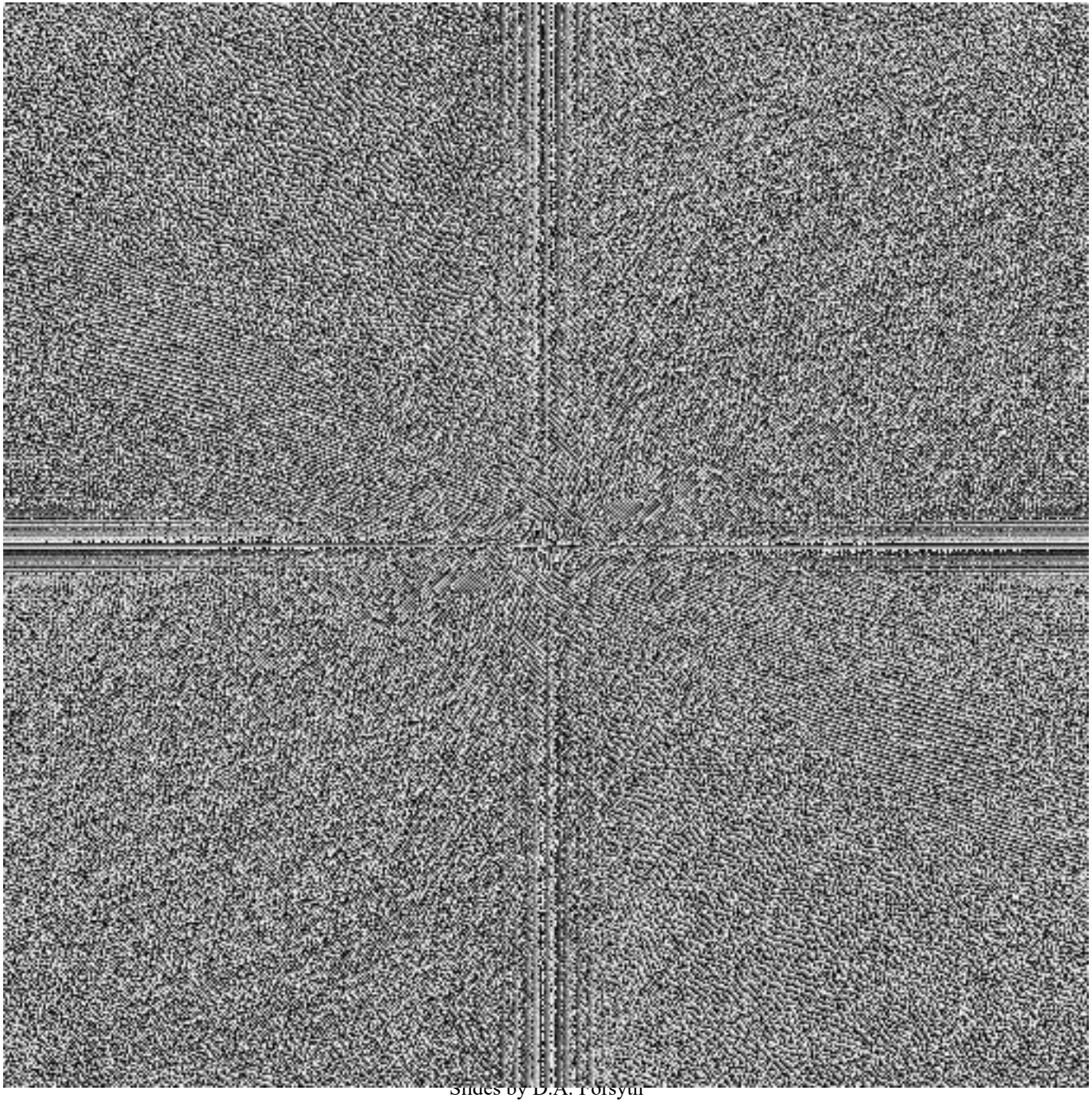
This is the phase transform of the cheetah pic



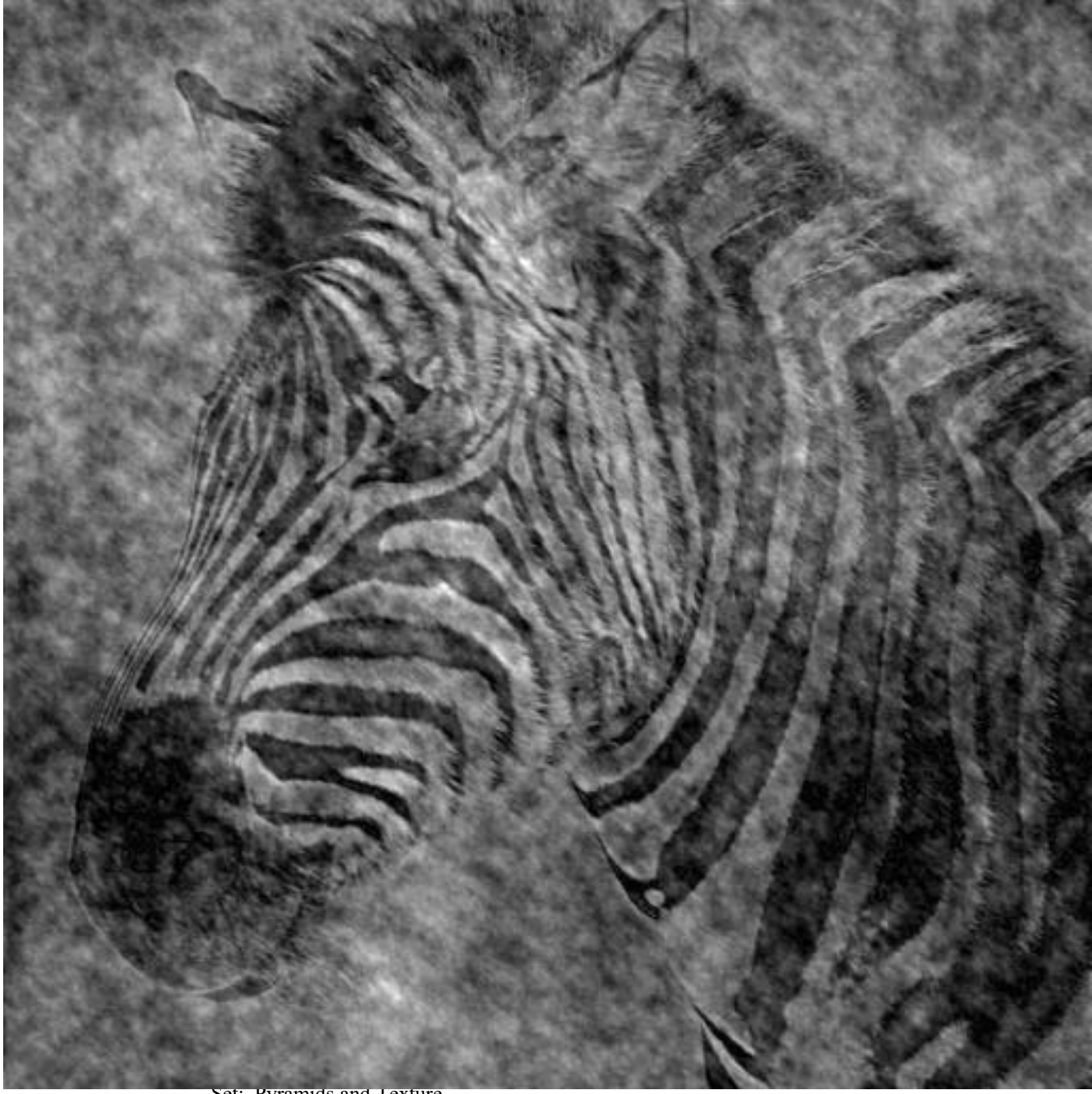
Slides by D.A. Forsyth

This is the magnitude transform of the zebra pic

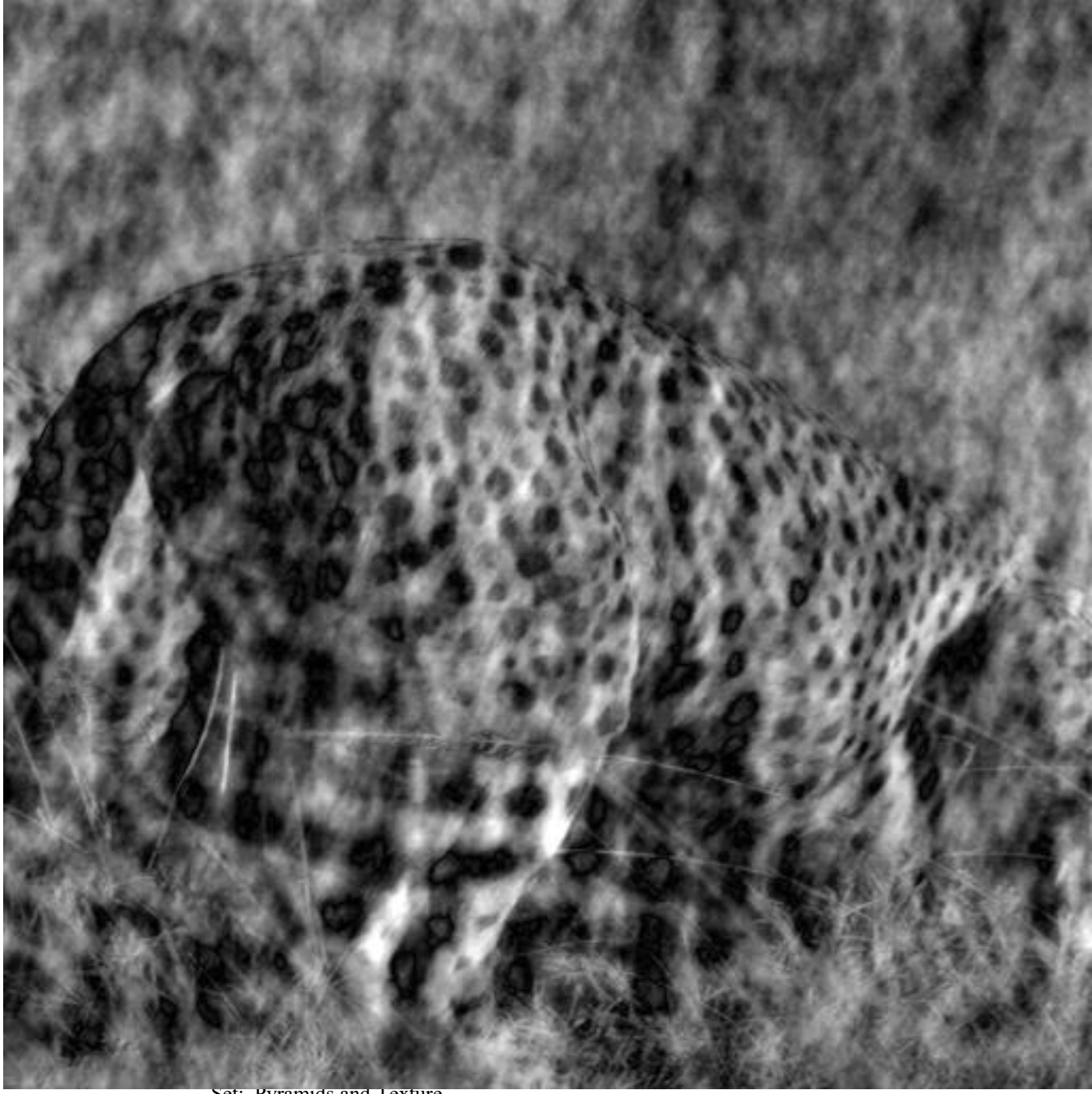
This is the phase transform of the zebra pic



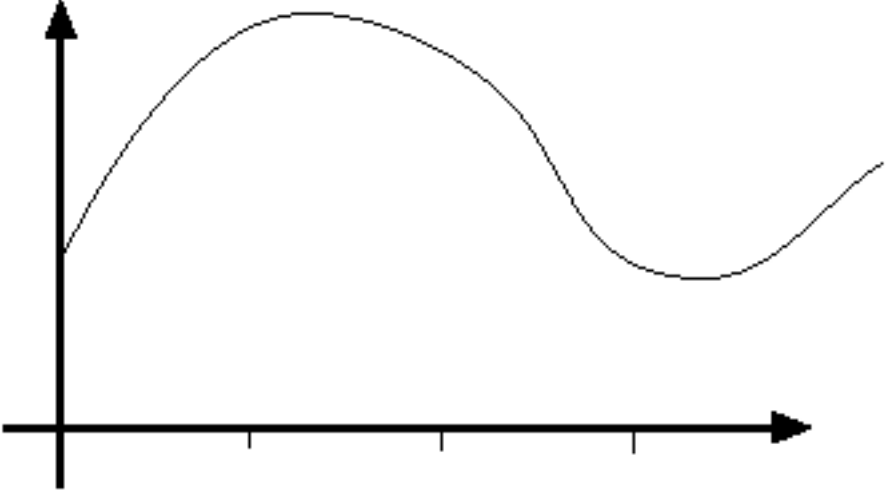
Reconstruction with zebra phase, cheetah magnitude



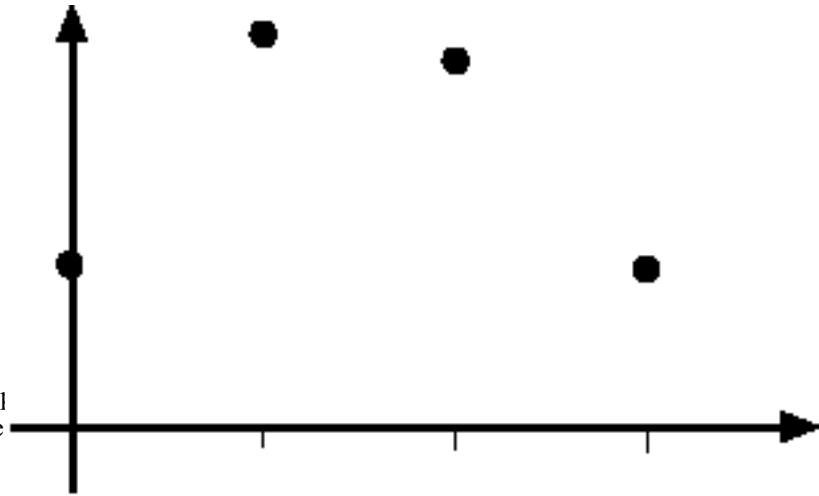
Reconstruction with cheetah phase, zebra magnitude

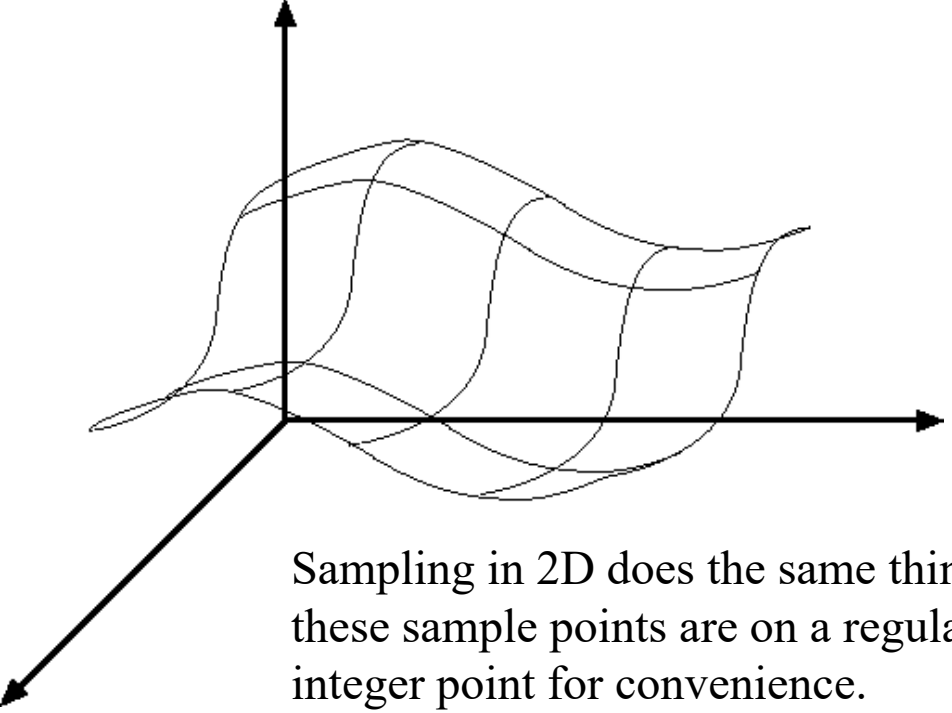


Sampling and Aliasing

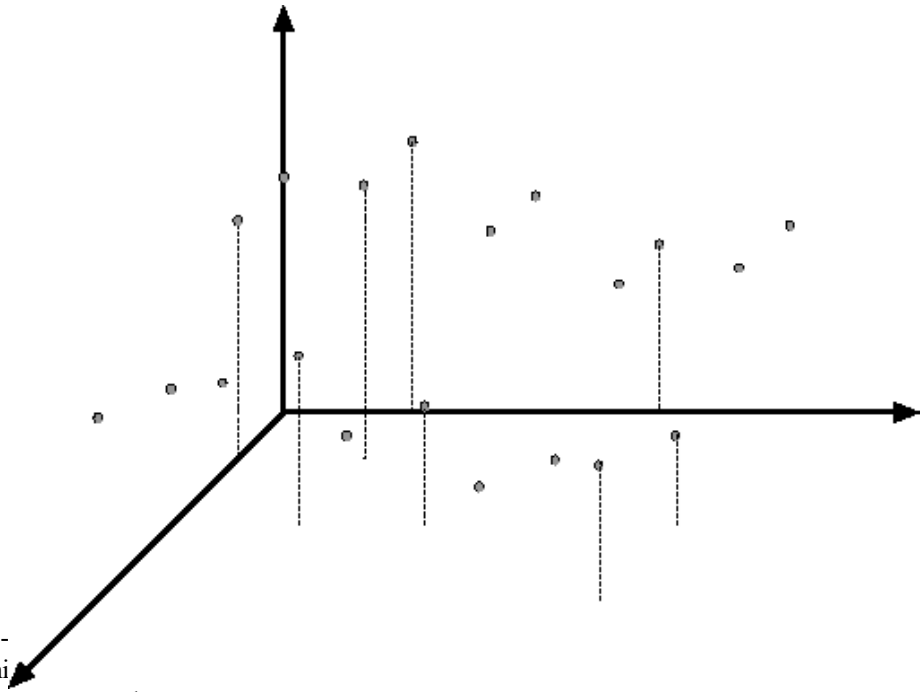


Sampling in 1D takes a continuous function and replaces it with a vector of values, consisting of the function's values at a set of sample points. We'll assume that these sample points are on a regular grid, and can place one at each integer for convenience.





Sampling in 2D does the same thing, only in 2D. We'll assume that these sample points are on a regular grid, and can place one at each integer point for convenience.



A continuous model for a sampled function

- We want to be able to approximate integrals sensibly
- Leads to
  - the delta function
  - model on right

$$\text{Sample}_{2D}\big(f(x,y)\big) = \sum_{i=-\infty}^{\infty}\sum_{i=-\infty}^{\infty} f(x,y)\delta(x-i,y-j)$$

$$= f(x,y)\sum_{i=-\infty}^{\infty}\sum_{i=-\infty}^{\infty} \delta(x-i,y-j)$$

The Fourier transform of a sampled signal

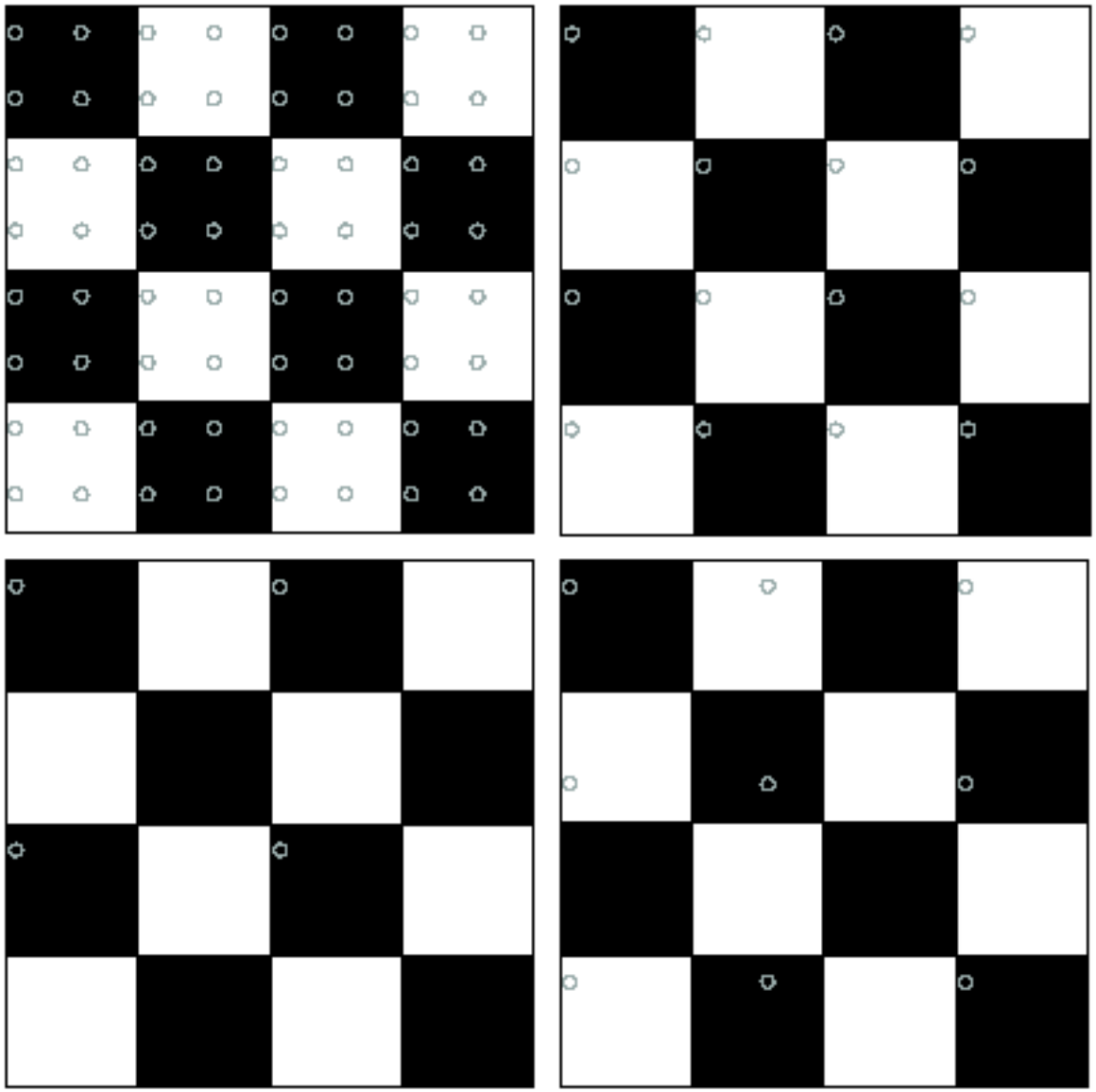
$$F\left(\text{Sample}_{2\text{D}}\left(f(x,y)\right)\right) = F\left(f(x,y)\sum_{i=-\infty}^{\infty}\sum_{i=-\infty}^{\infty}\delta(x-i,y-j)\right)$$

$$= F\left(f(x,y)\right) ** F\left(\sum_{i=-\infty}^{\infty}\sum_{i=-\infty}^{\infty}\delta(x-i,y-j)\right)$$

$$= \sum_{i=-\infty}^{\infty}\sum_{j=-\infty}^{\infty}F(u-i,v-j)$$

Aliasing

- Can’t shrink an image by taking every second pixel
- If we do, characteristic errors appear
  - In the next few slides
  - Typically, small phenomena look bigger; fast phenomena can look slower
  - Common phenomenon
    - Wagon wheels rolling the wrong way in movies
    - Checkerboards misrepresented in ray tracing
    - Striped shirts look funny on colour television



Resample the checkerboard by taking one sample at each circle. In the case of the top left board, new representation is reasonable.
Top right also yields a reasonable representation.
Bottom left is all black (dubious) and bottom right has checks that are too big.

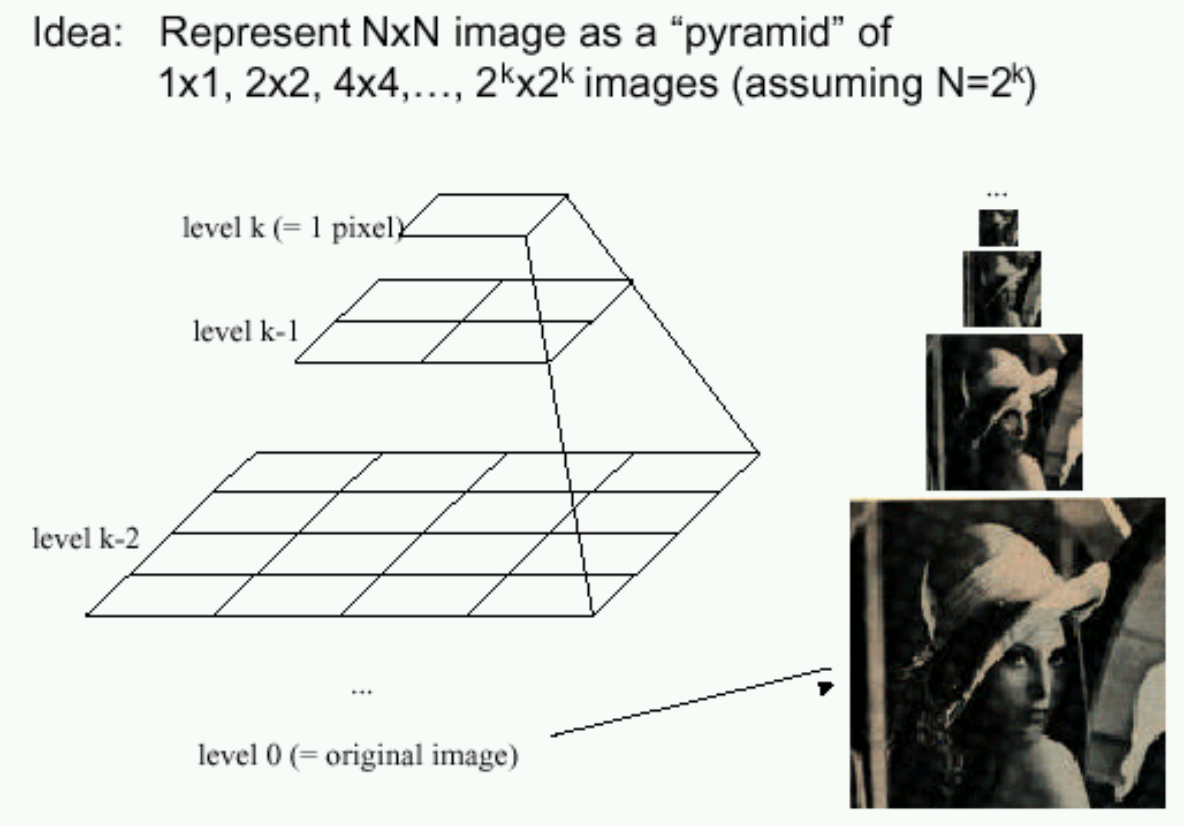
Smoothing as low-pass filtering

- The message of the FT is that high frequencies lead to trouble with sampling.
- Solution: suppress high frequencies before sampling
  - multiply the FT of the signal with something that suppresses high frequencies
  - or convolve with a low-pass filter
- A filter whose FT is a box is bad, because the filter kernel has infinite support
- Common solution: use a Gaussian
  - multiplying FT by Gaussian is equivalent to convolving image with Gaussian.

The Gaussian pyramid

- Smooth with gaussians, because
  - a gaussian*gaussian=another gaussian
- Synthesis
  - smooth and sample
- Analysis
  - take the top image

Gaussian Pyramid



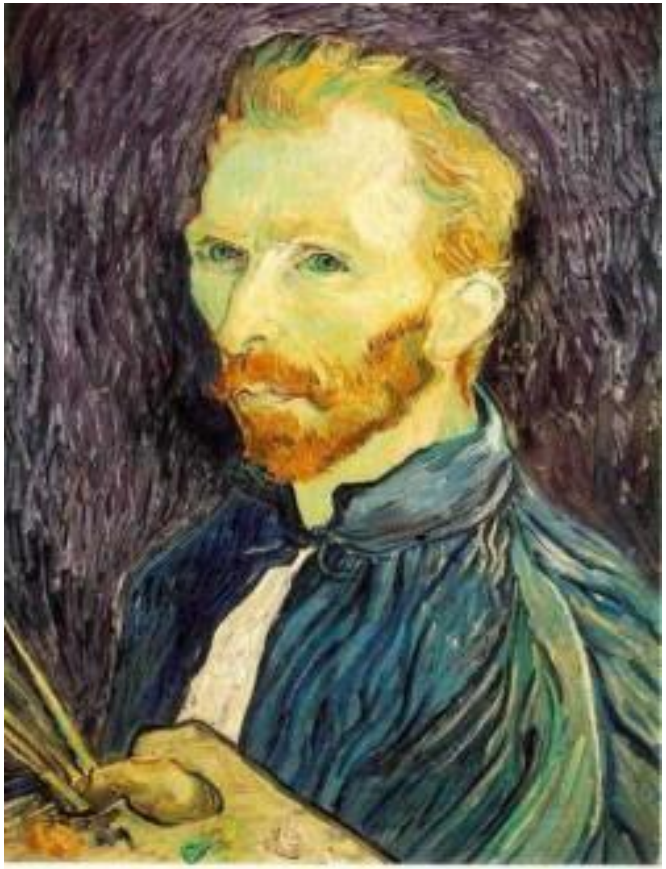
Known as a **Gaussian Pyramid** [Burt and Adelson, 1983]

- In computer graphics, a *mip map* [Williams, 1983]
- A precursor to *wavelet transform*

Gaussian Pyramids have all sorts of applications in computer vision

512 256 128 64 32 16 8

Subsampling with Gaussian pre-filtering



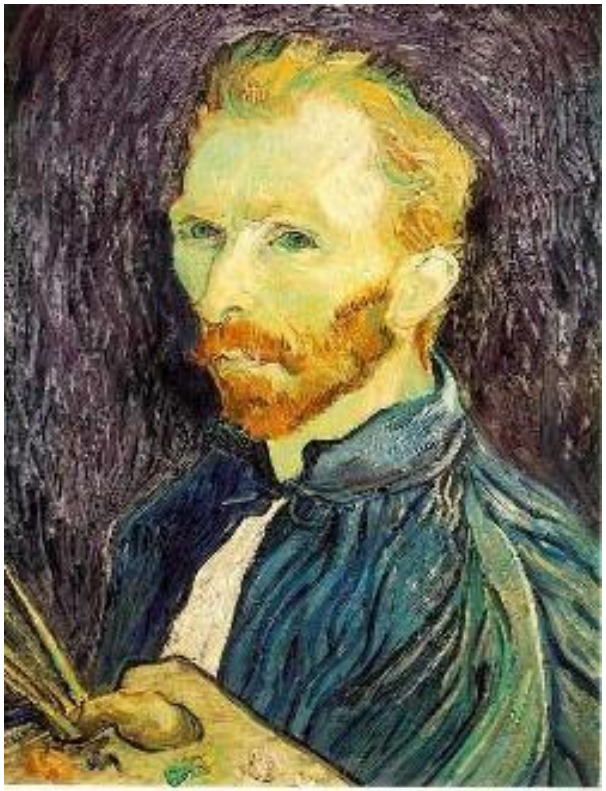
Gaussian 1/2

G 1/4

G 1/8

Filter the image, *then* subsample

Subsampling *without* pre-filtering

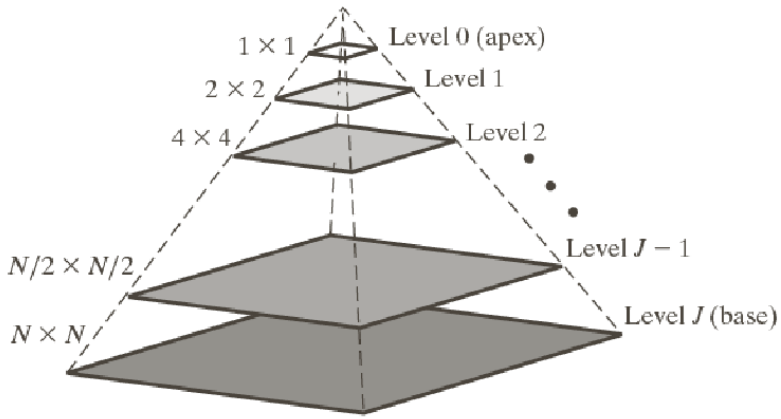


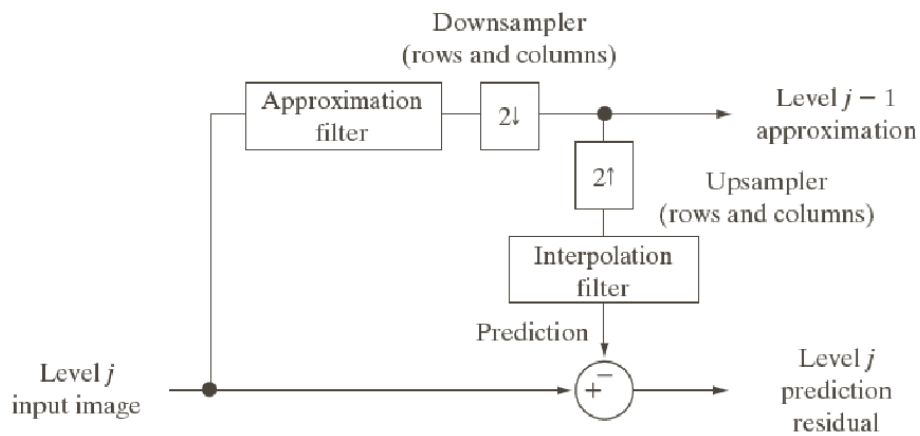
1/2

1/4 (2x zoom)

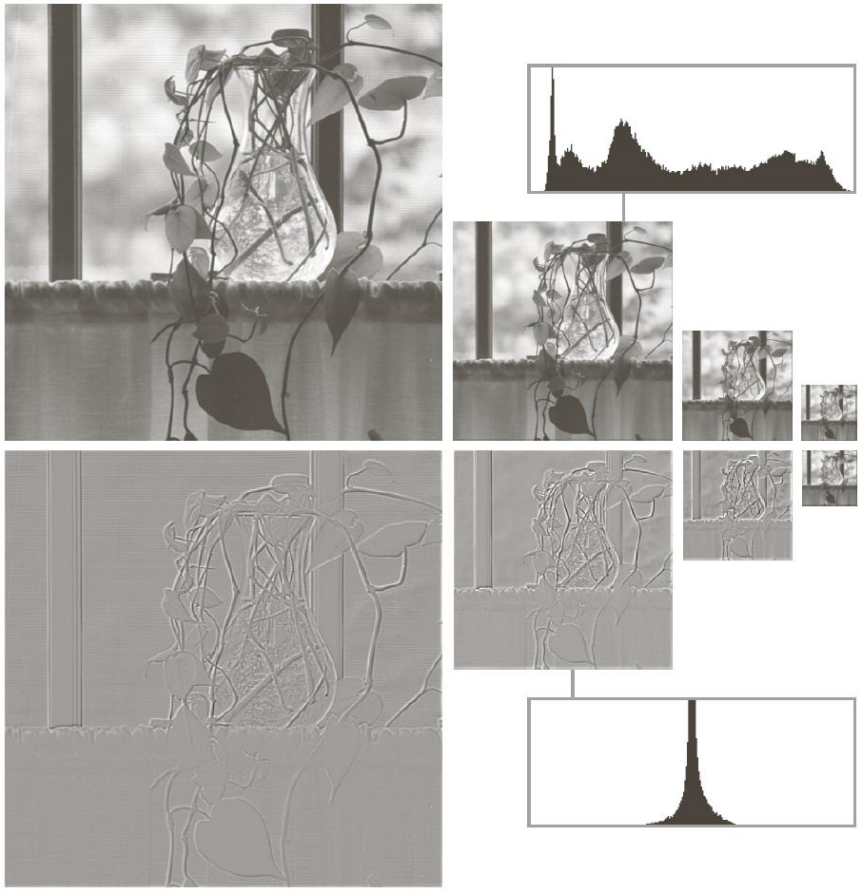
1/8 (4x zoom)

Image Pyramid – Approximation and Residual





Digital Image Processing – Gonzalez & Woods



a
b

FIGURE 7.3 Two image pyramids and their histograms: (a) an approximation pyramid; (b) a prediction residual pyramid.